

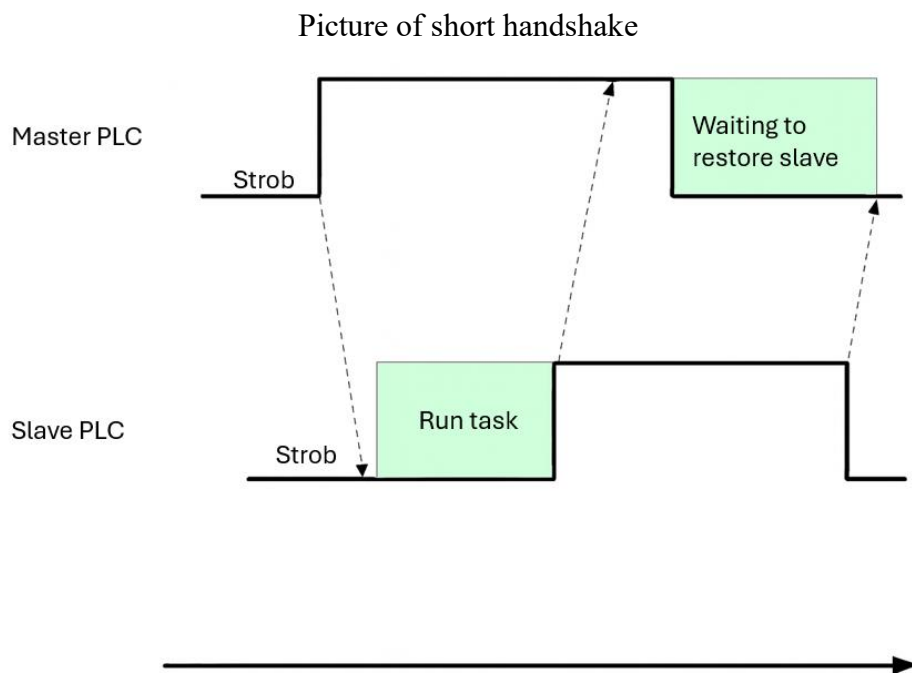
PLC Lab – Handshake

In this lab you will implement short and long handshake sequence control that interoperates with the provided Slave PLC project. For the examination, submit your code in a PDF format with pictures of code and comments on how it works on Canvas.

Handshake long sequence

1. **Task Data Transmission (Master → Slave)**
 - Master PLC sends task data to Slave PLC.
 - After sending, Master sets its strobe **High**.
2. **Acknowledgement of Task Reception (Slave → Master)**
 - Slave receives the task data and Master's strobe = High.
 - Slave sets its own strobe **High** to acknowledge.
3. **Konfirmation (Master → Slave)**
 - Master detects Slave strobe = **High**.
 - Master resets its strobe to **Low**.
4. **Task Exekution (Slave)**
 - Slave detects Master strobe = **Low**.
 - Slave resets its own strobe = **Low**.
 - Slave executes the assigned task.
5. **Completion Rapport (Slave → Master)**
 - When finished, Slave sends task result data to Master.
 - Slave sets its strobe **High**.
6. **Resultat Acknowledgement (Master → Slave)**
 - Master receives result data and Slave's strobe = **High**.
 - If data is correct, Master sets its strobe **High**.
7. **Final Konfirmation (Slave → Master)**
 - Slave detects Master strobe = **High**.
 - Slave resets its strobe = Low, enters wait state.

This sequence is essentially a **short handshake protocol** with single acknowledgements, ensuring that the task is executed and confirmed in a simplified manner. In this task, no actual data transfer will be carried out. The focus is solely on verifying the correct switching of the strobe signals between high and low on the PLCs.



Requirements

1. Start conditions
 - The sequence starts when the user presses the short or long start button in the HMI.
 - Starting initializes the handshake and triggers the car driving in the Slave PLC.
2. Sequence indication on HMI
 - While a sequence is running, lamps on the HMI must turn on/off to indicate activity.
 - The current step in the sequence must be indicated by the corresponding light.
3. Ready indication
 - Add a blinking light that clearly shows when the program is ready to run again.
4. Runtime behaviour
 - The program must be able to run indefinitely.
 - It must be possible to switch between short and long handshake between runs.

Design & implementation guidelines

- Structure your solution into appropriate programs and function blocks (IEC 61131-3).
- Reusable tasks (e.g., communication/handshake) should be implemented as function blocks.
- Do not mix unrelated functionality inside a single POU or FB.
- Aim for readability, modularity, and clear interfaces.
- No unnecessary global variables.
- Follow the naming standard provided in the course.

Verifikation

- Execute the entire project and verify that the handshake sequences match the Slave PLC behaviour.
- Debug and clean up your code before sending it in.